

(a) Campaign planning and execution preparation

Human scientist



Available tools



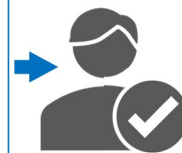
Conduct a montmorillonite water-vapor desorption campaign along RH $0.9 \rightarrow 0.3 \rightarrow 0.1$ to compare cation and layer-charge effects through water partitioning and basal-spacing responses.

Reasoning agent



Campaign planning

- 5 molecular systems with cation and layer-charge variations
- RH path: $0.9 \rightarrow 0.3 \rightarrow 0.1$
- Target observables: water content (interlayer and external) and basal spacing
- Stateful execution strategy and stopping criteria



Approved campaign

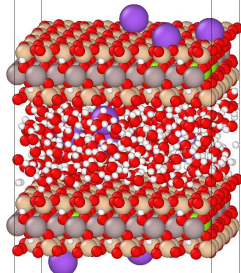
(Executable specification)

- LAMMPS (simulation execution)
- Force field files (Clayff, SPC/E)
- Python workflow scripts
- ClayCode (model generation)
- Previous templates

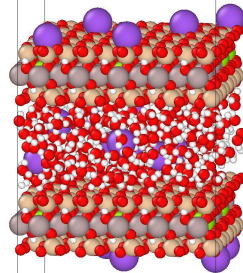
(b) Prepared molecular systems

○ H ● O ● Al ● Si ● Mg ● Na ● Ca ● K

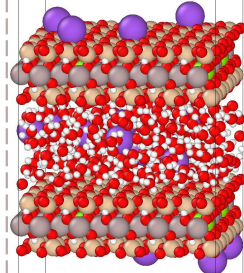
Layer-charge (LC) comparison with Na fixed



Na-LC0.3

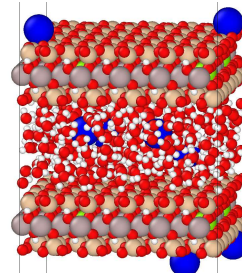


Na-LC0.5

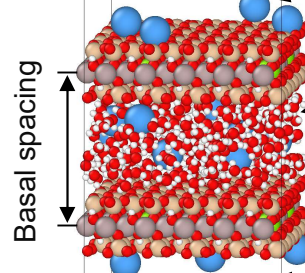


Na-LC0.4

Cation comparison at LC = 0.4



Ca-LC0.4



K-LC0.4

Basal spacing

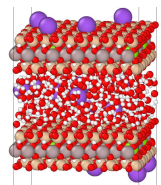
External space

Interlayer space

External space

(c) Stateful RH-dependent simulation campaign (per molecular system)

Initial model



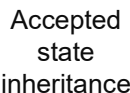
Initial equilibrium under NVT

State 1 (RH=0.9)

Run segment

Analyze outputs

Continue or accept

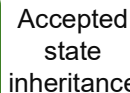


State 2 (RH=0.3)

Run segment

Analyze outputs

Continue or accept



State 3 (RH=0.1)

Run segment

Analyze outputs

Continue or accept

(d) Simulation outputs

